



USER MANUAL

Altura Rally Computer

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The Altura Rally Computer is designed exclusively for motorsport and competitive use. It is not approved for road use under standard vehicle regulations. The authors accept no liability for consequences arising from misuse.

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1.0 Introduction

The **Altura Rally Computer** is a GPS odometer and speedometer built for competitive stage rally use. It measures distance and speed using a u-blox NEO-6M GPS module — no wheel sensor required. The device provides precise distance measurement, live speed readout, GPS-synchronized time display, and detailed signal diagnostics on a compact 320×240 touchscreen.

Feature	Detail
GPS module	u-blox NEO-6M, 5 Hz update rate
Distance units	Miles
Display	320×240 touchscreen — fully touch-driven, no physical buttons
Saved settings	Survive power loss automatically
Visual themes	17 selectable colour schemes
Display fonts	8 selectable numeric fonts

i **For competitive use:** Keep the device powered continuously during events. Cold GPS starts can take up to three minutes. A recent warm start locks in under 30 seconds. Avoid power-cycling between recce and stages.

2.0 System Overview

The Altura runs two parallel processes that share data through a protected handoff, ensuring neither process can corrupt the other's work.

Process	Task	Responsibilities
GPS	Background	Receives GPS data, applies motion filters, accumulates distance
Display	Foreground	Updates screen every 100 ms, handles touch, runs animations

Distance Accumulation — Five Protection Layers

A GPS reading only contributes to your distance total if it passes all five checks below. Each layer guards against a different failure mode.

Check	Threshold	Protects Against
Valid location fix	GPS must report a confirmed position	Phantom readings during satellite search
Blackout recovery	Previous reading also had a valid fix	Distance jumps when signal returns
Minimum speed	Above 1.9 mph (approx. 3 km/h)	Stationary GPS noise accumulation
Speed change limit	Under 124 mph change between readings	Sudden GPS speed glitches
Per-reading distance cap	Under 1.0 mile per GPS update	Coordinate corruption jumps

3.0 First Boot & GPS Acquisition

On power-up, the Altura shows its boot splash for approximately 3.5 seconds, then transitions to the main screen. GPS acquisition begins automatically. Watch the three indicator lights in the top-left of the main screen.

Indicator	Colour / Pattern	Meaning
GPS	Grey, steady	No fix — fewer than 3 satellites
GPS	Green, blinking	Acquiring — satellites visible, accuracy still improving
GPS	Green, steady	Locked — ready to count distance
SATS	Grey, steady	Fewer than 3 satellites visible
SATS	Amber, blinking	2D fix — exactly 3 satellites (limited accuracy)
SATS	Green, steady	3D fix — 4 or more satellites (full accuracy)
HDOP	Grey, steady	No fix
HDOP	Green, steady	Good signal quality (HDOP under 2.5)
HDOP	Amber, blinking	Reduced quality (HDOP 2.5 – 3.5)
HDOP	Red, fast blinking	Poor signal quality (HDOP over 3.5)

i **Before a stage:** Wait for GPS steady green before the start line. Entering dense forest canopy will reduce accuracy — this is a GPS signal limitation, not a device fault.

i **Cold start:** If the device has been off for several hours or moved a long distance, acquisition can take one to three minutes. Keep it powered between recce runs and competitive stages when possible.

4.0 Core Concepts

4.1 Pending / Confirm Pattern

The Clock, Theme, and Font screens all use a two-step model: adjust a *preview* value first, then confirm or discard it. Nothing is ever saved unless you explicitly tap CONFIRM.

Step	Action	What Happens
1	Open screen	Current saved value loaded as the starting point
2	Adjust	Display updates immediately — live preview of the change
3a	Tap CONFIRM	Change saved and applied permanently. Returns to Settings.
3b	Tap BACK	Change discarded. Saved value unchanged. Returns to Settings.

4.2 Freeze Display (FRZ)

The FRZ button pauses the speed, total distance, and trip distance readouts. **Distance continues accumulating in the background** while frozen — un-freezing immediately shows the current accumulated values.

Behaviour	Detail
Button appearance	Turns red and blinks every half-second when active
Button label	Changes from FRZ to RES (resume) while frozen
Auto-release	Automatically releases after 60 seconds
Background counting	GPS distance accumulation continues normally
To un-freeze	Tap the red RES button, or wait 60 seconds

i *In a stage: Use FRZ to hold a reference reading while discussing a junction. The odometer keeps running and shows the correct total the moment you un-freeze.*

4.3 Odometer System

Counter	Label	Saved?	Reset With	Notes
Total distance	TOTAL	Yes — survives power loss	Long-press CLR ODO (~1.8 s)	Also resets average and peak speed
Trip distance	TRIP	No — clears on reboot	Long-press CLR INT (~1.8 s)	Use for stage-by-stage counting

Both CLR buttons require a sustained hold of about 1.8 seconds. The button darkens during the hold. Releasing early cancels the action — nothing is cleared unless you hold through the full animation.

4.4 Speed View Cycling

Tap anywhere in the large speed display area to cycle through three views. The small label below the number shows which view is active.

Label	Shows
mph	Current GPS speed (default)
avg	Average speed since the first GPS fix of this session
max	Highest speed recorded this session

Average and peak speed reset when you clear the total odometer. Tapping the speed area has no effect while the display is frozen.

4.5 Auto-Return

If no touch input is detected for 30 seconds on any settings screen, the device automatically returns to the main screen, preventing a settings screen from staying open mid-stage.

5.0 Navigation Model

All transitions use slide or fade animations. Going deeper slides left; going back slides right.

From	Action	Direction	To
Main Screen	Tap MENU	Slide left	Settings Menu
Settings Menu	Tap BACK	Slide right	Main Screen
Settings Menu	Tap GPS tile	Slide left	GPS Diagnostics Screen
Settings Menu	Tap CLOCK tile	Slide left	Clock Screen
Settings Menu	Tap THEME tile	Slide left	Theme Screen
Settings Menu	Tap FONT tile	Slide left	Font Screen
Settings Menu	Tap RESET tile	Slide left	Factory Reset Screen
Any settings screen	Tap BACK	Slide right	Settings Menu
Clock / Theme / Font	Tap CONFIRM	Slide right	Settings Menu (change saved)
Any screen	30 s no activity	Fade	Main Screen

6.0 Screen Guide

Each screen below is shown at its correct aspect ratio using the default Altura colour scheme. The Boot Splash, Main Screen, and Settings Menu use actual device screenshots.

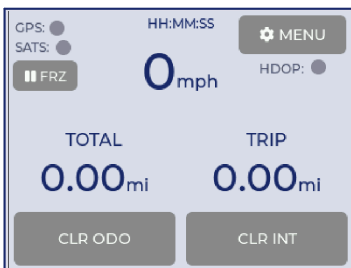
6.1 Boot Splash Screen



Boot Splash Screen

Shown for approximately 3.5 seconds on power-up.
Displays the Altura logo and firmware version.
GPS begins acquiring signal during this screen.
Transitions automatically to the Main Screen.
No user interaction required or possible.

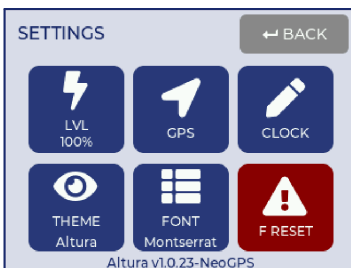
6.2 Main Screen



Main Screen

GPS / SATS indicators: Coloured dots show fix status.
Clock: GPS time adjusted to your UTC offset.
MENU: Opens the Settings Menu.
FRZ: Freezes the readouts (distance still accumulates).
HDOP: Real-time GPS signal quality indicator.
Speed: Tap to cycle current / average / max.
TOTAL and TRIP: Distance counters in miles.
CLR ODO / CLR INT: Hold ~1.8 s to reset.

6.3 Settings Menu



Settings Menu

6 tiles in a 3×2 grid. Tap any tile to open that setting.
BRIGHT: Cycles brightness 25% / 50% / 75% / 100%.
GPS: Opens the GPS Diagnostics Screen.
CLOCK: Opens the Clock Screen to set UTC offset.
THEME: Opens the Theme Screen (17 colour schemes).
FONT: Opens the Font Screen (8 numeric fonts).
RESET: Always dark red — opens factory reset confirmation.

6.4 GPS Diagnostics Screen

GPS DIAGNOSTICS		BACK
Latitude	46.412345	
Longitude	-87.654321	
Altitude	1204 ft	
HDOP	1.8	
Fix Quality	GOOD	
Satellites	7	
Rej. Steps	0	
Rej. Sanity	0	
Blackouts	3	

GPS Diagnostics Screen

- Live diagnostic table. Scroll up/down to view all rows.
- Latitude / Longitude:** Current position to 6 decimal places.
- Altitude:** Current altitude in feet.
- HDOP:** Row highlights green (good), amber, or red (poor).
- Fix Quality:** EXCL / GOOD / POOR / NO FIX.
- Satellites:** Number of satellites used.
- Rej. Steps / Sanity:** Readings blocked by spike filters.
- Blackouts:** Fix interruption count (normal in dense canopy).

6.5 Clock Screen

CLOCK		CONFIRM	BACK
14:32:07			
UTC-5			
- +			
Hold for fast repeat			

Clock Screen

- Sets your local UTC time zone offset.
- Uses the Pending / Confirm pattern (Section 4.1).
- Large clock previews the new time immediately as you adjust.
- Main screen clock stays at the old time until you confirm.
- / + buttons:** Adjust offset -14 to +14 hours. Hold to repeat.
- CONFIRM:** Saves and returns to Settings.
- BACK:** Discards changes and returns to Settings.
- Factory default: UTC-5 (Eastern Standard Time).

6.6 Theme Screen

THEME SELECT		CONFIRM	BACK
Altura			
Night Stage			
Day Stage			
Stealth			
Cobalt			
...17 themes, scroll up/down			

Theme Screen

- 17 colour themes as a scrollable list.
- Each row shows five colour swatches for that theme.
- Current theme pre-highlighted when you open.
- Tap a row to stage it as your pending choice.
- CONFIRM:** Applies to all screens and saves.
- BACK:** Discards and restores the current theme.
- The Factory Reset tile stays dark red on every theme.

6.7 Font Screen

FONT SELECT		CONFIRM	BACK
Montserrat 4 2 0 6 9			
B612 4 2 0 6 9			
DIN1451 4 2 0 6 9			
DSEG14 4 2 0 6 9			
DSEG7 4 2 0 6 9			
LCD 4 2 0 6 9			
...8 fonts, scroll up/down			

Font Screen

- 8 numeric fonts as a scrollable list.
- Each row previews that font using sample numbers.
- Applies to speed, total, trip, and clock readouts.
- All menus and button labels always use the default font.
- CONFIRM:** Applies and saves.
- BACK:** Discards the selection.

6.8 Factory Reset Screen



Factory Reset Screen

CANCEL: Returns to Settings. Nothing is changed.

RESET (dark red): Immediately erases all saved data.

Resets: odometer, trip, avg/peak speed, all counters.

Resets all settings to factory defaults.

Shows confirmation for one second, then returns to main screen.

This action cannot be undone.

7.0 GPS Signal Indicators

Three round indicator lights appear in the top-left of the main screen. Colour is the only signal — the lights are always on; muted grey means no data.

GPS Indicator — overall fix quality

Colour	Pattern	Condition	Meaning
Grey	Steady	Fewer than 3 satellites	No fix
Green	Blinking (0.5 s)	3+ satellites, HDOP above 2.5	Acquiring signal
Green	Steady	3+ satellites, HDOP 2.5 or below	Locked and ready

SATS Indicator — fix type

Colour	Pattern	Condition	Meaning
Grey	Steady	Fewer than 3 satellites	No position fix
Amber	Blinking (0.8 s)	Exactly 3 satellites	2D fix — limited accuracy
Green	Steady	4 or more satellites	3D fix — full accuracy

HDOP Indicator — horizontal signal accuracy

Colour	Pattern	Threshold	Meaning
Grey	Steady	No fix	No satellite fix
Green	Steady	HDOP 2.5 or below	Good signal quality
Amber	Blinking (0.8 s)	HDOP 2.5 – 3.5	Reduced signal quality
Red	Fast blink (0.25 s)	HDOP above 3.5	Poor signal quality

8.0 GPS Quality & Accuracy

Fix Quality Labels

Label	HDOP Range	Description
EXCL	Below 1.5	Excellent — best possible conditions
GOOD	1.5 – 2.5	Good — normal operating range
POOR	Above 2.5	Degraded — weak satellite geometry
NO FIX	n/a	Fewer than 3 satellites visible

Dense Canopy Performance

On stages with dense forest canopy, the GPS module periodically loses position lock. When the fix returns, counting resumes from the new position — distance traveled during the blackout is not recovered. This is a GPS hardware limitation.

Terrain	Typical Accuracy
Open stages / highway roads	Approximately 0.1% error
Dense canopy stages	4–8% undercount possible

GPS Module Settings (applied automatically at each startup)

Setting	Value	Purpose
Update rate	5 Hz	Five position readings per second
Active sentences	Position and speed only	Eliminates unnecessary data overhead
Motion model	Automotive	Optimised for ground vehicle movement
Stationary filter	Active	Suppresses phantom movement when parked
SBAS corrections	Enabled (WAAS)	Improves absolute position accuracy
Horizon mask	10 degrees	Filters low-angle, high-interference satellites

9.0 Settings & Saved Data

All settings survive power loss. Values are validated on startup — corrupted values are reset to factory defaults automatically.

Setting	Default	Notes
Total odometer	0.00 mi	Saved every mile and on every manual reset
UTC time offset	UTC-5	Range: -14 to +14 hours
Screen brightness	100%	4 levels: 25% / 50% / 75% / 100%
Visual theme	Altura	17 themes available
Display font	Montserrat	8 fonts available

i **Odometer saves:** The total distance is written to storage each time one mile accumulates, and immediately whenever you clear it or perform a factory reset. In the event of a sudden power cut, up to one mile of distance could be lost — the trip counter is not saved across power cycles.

10.0 Troubleshooting

All GPS indicators are grey after power-up

Normal for the first 30–90 seconds (warm start) or up to three minutes (cold start). Move to a clear sky location if acquisition exceeds five minutes.

Distance is counting while the vehicle is stationary

A minimum speed of about 1.9 mph is required before any distance is added. The built-in stationary filter and speed gate together keep this negligible under normal conditions.

TRIP is correct but TOTAL seems wrong

CLR ODO and CLR INT are independent buttons with independent functions. Both require a firm 1.8-second hold to activate.

I accidentally opened a settings screen during a stage

Tap BACK. Nothing changes unless you tap CONFIRM.

Speed display shows "avg" or "max" instead of current speed

The speed readout area was tapped. Tap it again until the label reads "mph".

The device returned to the main screen on its own

The 30-second inactivity auto-return triggered — this is intentional.

FRZ will not turn off

Tap the red blinking RES button. FRZ also releases automatically after 60 seconds.

The clock is showing the wrong time

Open Settings > Clock and adjust the UTC offset to match your time zone. Tap CONFIRM.

The odometer shows a value after a factory reset

Confirm the reset completed — the RESET COMPLETE message should have appeared. If the process was interrupted, perform the reset again.

"Rej. Sanity" is climbing on the GPS Diagnostics Screen

The speed-change filter is triggering repeatedly — usually heavy canopy or a blocked antenna. The odometer is protected from these bad readings.

End of Altura Rally Computer User Manual — v1.0